



Understanding the expressivity of multi-headed attention

Solving AI alignment by letting LLMs run wild on mech interp puzzles and formally verifying them.

Intro meeting · June 2026

Hi, I'm Karthik

Physics PhD working on **mechanistic interpretability** and **AI alignment**.

- From **IIT Madras** to a Physics PhD at the **University of Amsterdam**.
- Recent alignment research: **MATS 9.0** (Berkeley) and **MARS 3.0** (Cambridge).

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Mechanistic interpretability

“Reverse engineering neural networks, from the learned weights down to human-interpretable algorithms.”

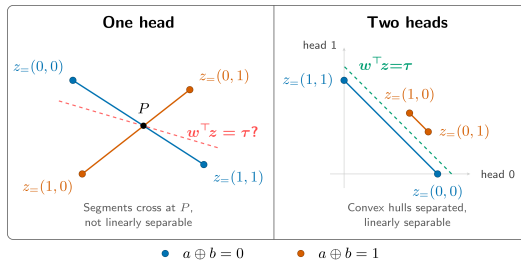
– a famous mechanistic interpretability researcher

- The most fundamental level of the AI alignment problem. Knowing what a model computes inside is the basis for **trusting and steering** it.
- Today this is mostly empirical. We rarely get **exact, provable** statements about what a component can represent.

Our angle: start from the **smallest questions we can answer exactly**.

A crisp instance: XOR in one attention layer

How many heads does a single attention layer need for XOR? **Exactly two**

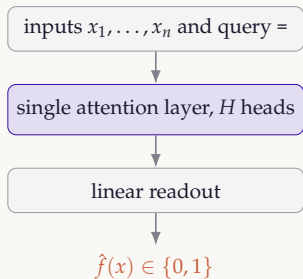


One attention head cannot compute XOR, but two can.

Figure: "How many attention heads do you need to do XOR?", LessWrong.

Scaling it up: from XOR to $H^*(f)$

$H^*(f)$: the **fewest attention heads** that compute a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, in a single attention layer with a linear readout.



- Proof of concept: $H^*(\text{parity}_n) = n$, while $H^*(\text{AND}) = H^*(\text{OR}) = 1$.
- A complexity measure **native to transformers**.

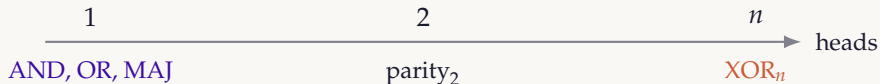
What we know so far

Lower bounds

$$\deg_{\pm}(f) \leq H^*(f)$$

Exact

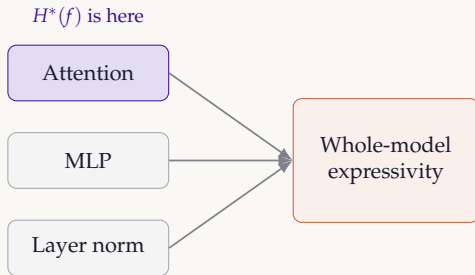
$$H^*(\text{XOR}_n) = n$$



We can bound many families, but no known invariant pins down $H^*(f)$ exactly. This is our first open problem.

The grand goal

- Attention heads are **one component**.
The grand goal is to characterize **each** transformer component, then chain them together.
- A compositional theory of **what transformers can represent**.
- $H^*(f)$ is the **first brick**.



Why now: LLMs change the economics of math

LLMs can do real math

GPT-5 produced solutions to open Erdős problems, several later checked in Lean (2025).

LLMs can write Lean

Galois, “*Claude Can (Sometimes) Prove It*”: real proofs by **directing a coding agent**, barely any hand written Lean.

Proving results at this scale is finally **within reach**.

Trust, but verify: the Mech Interp Dojo

- LLM drafts, Lean verifies. LLM proofs hallucinate; **Lean 4 + mathlib** checks every one and steers the proof generation.[†]
- Humans ask the right questions and design abstractions; agents do the proof engineering.

```
theorem xor_needs_two :  
  Hstar (xor 2) = 2 := by  
  -- formal proof  
  ✓ checked by Lean
```

[†] Even though this sounds like a neat plan, I have not yet seen verification help an LLM discover something it could not have found on its own. I am looking forward to seeing that happen here.

Deliverables

A library of lemmas

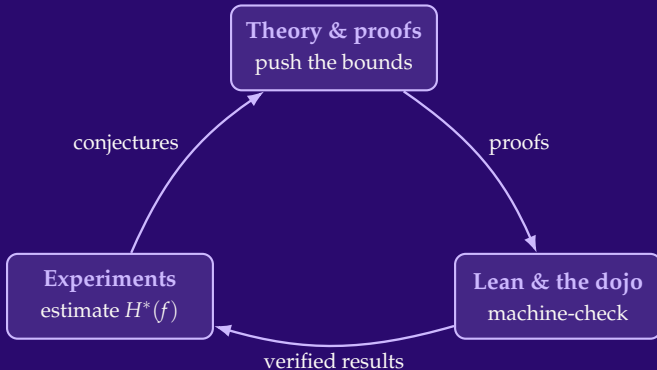
Machine-checked theorems on **what transformer components can and cannot represent**, from $H^*(f)$ and beyond.

The Mech Interp Dojo

An open source **Lean 4 + mathlib** (modular) codebase that can be used by other researchers formally verifying proofs for mech interp.

Where you come in

Three connected tracks, jump in anywhere.



Important links

Intro to the project

- **Preliminary result:** How many attention heads do you need to do XOR? (LessWrong)
- **Project proposal:** [proposal.pdf](#)
- **Code:** github.com/karthikviswanathan/how-many-attention-heads-xor
- **Team check-ins:** [shared doc](#)

Some questions

- How should we communicate? **Slack** or **Discord**?
- How often should we meet?
- LLM credits: should we apply to **Rapid Grants** (BlueDot Impact)?



Thank you!

Let's solve AI alignment, one verified proof at a time!